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Primary mercury gas standard for the calibration of mercury measurements

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ABSTRACT

Mercury poses one of the greatest current direct threats to human, animal and environmental health across the globe. Robust, defensible and traceable measurements of mercury concentrations are essential to underpin global efforts to reduce the concentration of mercury in the environment, meet the obligations of legislation and to protect human health. Currently, instruments for gaseous mercury concentration measurements are calibrated based on mercury vapour pressure equations (e.g., the Dumarey or the Huber equation). Currently these equations differ from each other by more than 7% at 20 °C. In this paper, the performed characterisation and implementation of mercury diffusion cells and the development of the first primary mercury gas standard show that metrological traceability to the International System of Units (SI) is possible for mercury concentrations. The primary mercury gas standard will provide a lower measurement uncertainty (between 1.8% and 3%) than can be obtained using the most commonly adopted mercury vapour pressure equations (5% up to 21%). Measurement methods have been developed for the SI traceable calibration of mercury analysers and mercury gas generators which have been used for the comparison between the primary mercury gas standard and the Dumarey equation. The developed primary mercury gas standard will contribute to more reliable mercury measurement results at emission and ambient air levels.

1. Introduction

Mercury is a trace component in all fossil fuels, whereas natural sources such as volcanoes are responsible for about half of the atmospheric mercury emissions. About 65% of the human-caused emissions is emitted by stationary combustion. Once it is released into the environment, mercury can circulate through air, soil, water and animals for thousands of years. The current levels of mercury in the atmosphere (2–4 ng m^{-3}) are up to 5 times above natural levels. Mercury is one of the top ten chemicals of major public health concern and a substance which disperses into and remains in ecosystems for decades, causing severe illness and intellectual impairment to exposed populations [1].

To protect the environment from the adverse effects of mercury, emissions are regulated by the Industrial Emissions Directive (IED) 2010/75/EU [2], the Air Quality Directive 2004/107/EC [3], the Waste Incineration Directive 2000/76/EC [4] and worldwide by the Minamata Convention [5]. At the moment it is not possible to defensibly assess mercury at relevant concentration levels as required by the European directives, because of a lack of underpinning metrology and validated methodologies for mercury concentrations in emission sources (μg m^{-3}) and the atmosphere (ng m^{-3}).

Nevertheless, requirements for metrological traceability of mercury measurement results from stationary sources do exist in the IED. Furthermore, ISO/IEC 17025 requires measurement results to be traceable to the International System of Units (SI) where possible [6]. Traceable methods and calibration standards for elemental mercury (Hg^0) are based upon mercury vapour pressure equations, e.g., the Dumarey equation or the more recent Huber equation, that currently differ from each other by more than 7% at 20 °C [7–9]. This discrepancy is of great concern as it hampers comparable and reliable measurement results for mercury concentrations in the atmosphere, emissions and natural gas. Calibration methods currently used in the field have relative uncertainties ranging from 5% up to 21% [9–13]. More recently the development of two analytical methods has been described for the National Institute of Standards and Technology (NIST) certification of a primary gas standard for elemental mercury [14].

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The last decade, considerable effort has been put into the development of a primary gas standard for mercury concentration [15]. The concentration realised by this standard can be computed from first principles (diffusion, gravimetry and volumetry in this instance [16]), thereby having the characteristics of a primary method of measurement [17–20].

Ent et al. described the principle for the gravimetric approach to provide SI traceability for concentration measurement results of mercury [15]. Using specially designed diffusion cells, elemental mercury is vaporized under well-controlled conditions (temperature, flow and pressure) according to ISO 6145-8 [16]. By weighing the diffusion cells at regular time intervals with a high-resolution balance, an accurate mercury diffusion, or mass flow rate, traceable to the SI, is obtained. After mixing the mass flow of mercury with accurately measured mass flows of nitrogen and air, a primary mercury gas standard has been obtained.

In this work, several improvements to the original design are presented, together with the results of the validation of the system. Diffusion cells, made from stainless steel instead of glass, and an advanced mercury loss weighing process, including a specially designed pressure chamber, have been developed. The mercury diffusion from the stainless steel diffusion cells has been fully characterised and implemented to obtain primary mercury gas standards for the range of 0.1–100 μg m^{-3}, with an expanded measurement uncertainty between 3% and 1.8%, which could not be obtained using glass diffusion cells [15]. Calibration methods currently used in the field based on mercury vapour pressure equations (e.g.; the Dumarey or the Huber equation) cannot be used to validate the primary mercury gas standard. Their agreement in predicting the mercury vapour pressure is insufficient to obtain the required uncertainty, as the discrepancy of more than 7% at 20 °C is too large [7–9]. A comparison between a calibration with the primary mercury gas standard and a calibration based on the Dumarey equation can however be used to determine the difference between them. The results of the primary standard characterisation and comparisons with the Dumarey equation are shown in this paper.

The developed primary mercury gas standard is unique in the world and measurement methods have been developed for the direct calibration of mercury analysers and the certification of mercury gas generators used in the field, as shown in this paper, but also for indirect calibration or certification via transfer standards [21]. These SI-traceable calibration methods are of fundamental importance to reduce the mercury burden on the environment, to comply with related regulation and to protect human health. Their use enables a harmonised and reliable way of monitoring claimed emission reductions and provides credible data underpinning the implementation of the Minamata convention.

2. Materials and methods

2.1. Stainless steel diffusion cells

In 2015 Ent et al. published the characterization of glass diffusion cells to obtain a gravimetric approach to provide SI traceability for mercury concentration measurement results [15]. These glass diffusion cells proved to give a stable diffusion rate of mercury and showed the potential to produce SI traceable measurement results. Nevertheless, the measurement uncertainty was too large [15]. Therefore, improved, more robust and less fragile diffusion cells have been manufactured from stainless steel instead of glass. To obtain several diffusion ranges different sizes were designed and produced out of solid stainless steel (AISI 316). The cells with a capillary of 33 mm in diameter were produced from an open tube of stainless steel. In the cells with a capillary of 1, 3 and 8 mm in diameter, the opening was created later by pressurised water drilling. The length to inner diameter ratio determines the diffusion which required 50 μm tolerances and a surface roughness of ≤0.4 Ra. The bottom disc is tungsten inert gas (TIG) welded to the capillary part with backing gas to prevent any oxidation during the welding process. To ensure an air tight cell, the cap contains a thin Viton 70^{o}Sh O-ring which is compressed by two firm springs.

After production the cells were cleaned and filled with approximately 2 mL of elemental mercury (electronic grade; Aldrich, product number 294594; Lot SLBC3181V) according to the procedures published by Ent et al. [15]. In this paper the results of the full characterisation of the cells with a capillary of Ø3 mm and Ø33 mm (Fig. 1) are presented. Three identical cells were made with a capillary of Ø3 mm and 50.5 mm in length (diffusion cells with numbers 18, 19 and 20) and Ø33 mm and 100 mm length (diffusion cells with numbers 34, 35 and 36). In the near future, diffusion cells with a smaller opening (Ø1 mm) will be characterised to obtain mercury concentrations in the range from 0.01 μg m^{-3} to 0.45 μg m^{-3}.

2.2. Pressure chamber

In order to realise SI traceability, the mercury diffusion rate is determined by weighing the diffusion cells at regular time intervals with a high-resolution and high accuracy balance. Before weighing the pressure inside the diffusion cell is adjusted to 102.00 kPa ± 0.01 kPa by use of a unique pressure chamber (Fig. 2). It is key to determine the mass of the cells at the same internal pressure. A difference of 0.1 kPa inside the cells gives rise to a difference of 5 μg per weighing while the smallest cells (Ø1 mm) only have a diffusion of 100 μg per year. Correcting for the difference in ambient pressure is difficult, but, more importantly, would result in a too large uncertainty source.

Before weighing, the diffusion cell is closed and placed inside the pressure chamber. The pressure chamber is developed to open the diffusion cell for a certain amount of time to adjust the pressure inside the diffusion cell to the desired pressure. The pressure chamber is equipped with a pressure sensor and an oxygen sensor. The oxygen sensor determines the oxygen concentration in the chamber. As a preventative measure it is important not to allow oxygen within the diffusion cell as elemental mercury can form oxidized mercury species which have a greater mass. This is an important factor when microgram quantities of mass difference are determined. The pressure chamber is connected to pure nitrogen for flushing and removing oxygen. When the oxygen concentration in the chamber is below 0.1% the diffusion cell is opened for one minute to adjust the pressure in the diffusion cell to the chamber's pressure.

2.3. Balance

After setting the pressure inside the Ø3 mm diffusion cells to 102.00 ± 0.01 kPa they are placed inside the AX1006 mass comparator (Mettler, Switzerland) with a Mettler AT1005 balance (Mettler, Switzerland), inside an airtight container under air [15]. The setup has a resolution of 1 μg and repeatability of 2 μg. The temperature in the weighing chamber is determined with two thermistors and the pressure is controlled at 102.00 ± 0.05 kPa using a pressure controller.

The Ø33 mm cells are much bigger in size, which does not only lead to a mercury diffusion rate which is approximately 50 times bigger, but also make the cells unsuitable for pressure correction in the pressure chamber and weighing with the AX1006 mass comparator. The Ø33 mm cells are weighed against a reference weight on a Sartorius CC500 analytical balance (Sartorius, Germany) when the ambient pressure is between 101.5 kPa and 102.0 kPa, to prevent extremes in ambient pressure. The Sartorius CC500 balance has a capacity resolution of 505 g × 0.01 mg. Due to the 50 times higher diffusion rate the pressure is a less important factor compared to the diffusion rates obtained with smaller cells.

Weighing of a reference weight and the Ø3 mm or Ø33 mm cells was performed through metrological gravimetric principles by substitution measurements, to determine the mass difference between each diffusion cell and the reference weight. The reference weights used conform to OIML class E1 [22]. The reference weight used during the weighing of

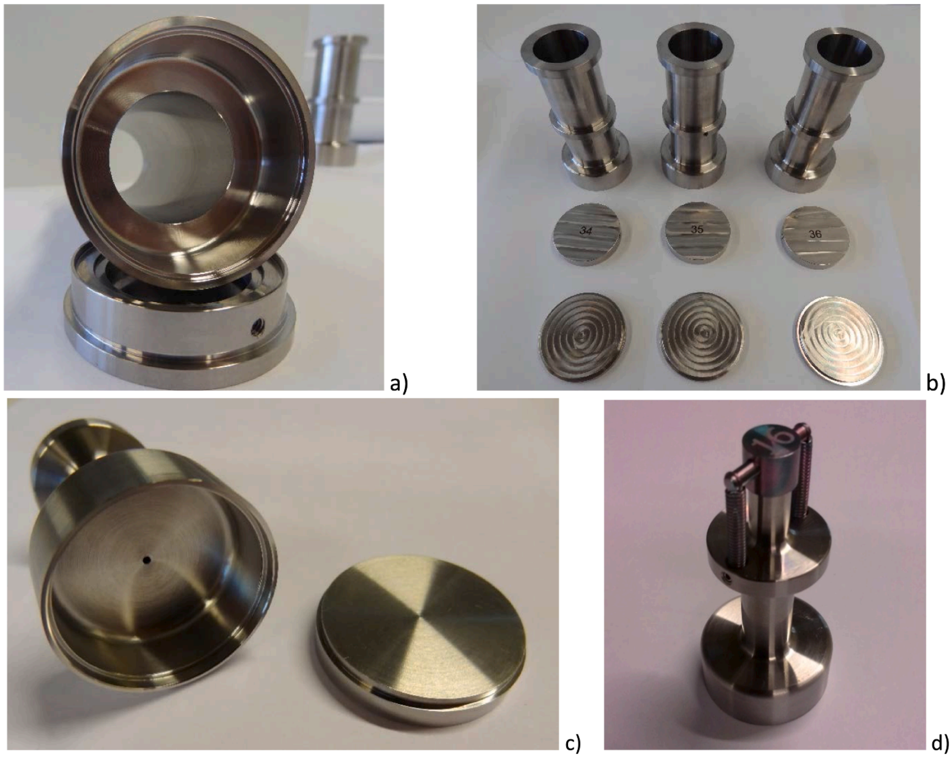

Fig. 1. (a) Ø33 mm stainless steel diffusion cells, the capillary not assembled to the bottom plate yet, (b) the three identical capillaries, bottom plates and caps of the Ø33 mm stainless steel diffusion cells, (c) Ø1 mm stainless steel diffusion cell, the capillary not assembled to the bottom plate yet and (d) assembled Ø1 mm stainless steel diffusion cell (picture is enlarged).

Fig. 2. Pressure chamber.

the Ø3 mm diffusion cells has been specially made with a weight of 120 g.

2.4. Dynamic mercury gas generator

Instrumentation was developed for the dynamic mercury gas generation, to house the diffusion cells and dilute the mercury mass flow from the open diffusion cells to the desired concentration. The dynamic mercury gas generator was described by Ent et al. [15]. As described in that publication the interior (diffusion chamber) of the separate dynamic mercury generators was SilcoNert 1000 (SilcoTek, USA) treated to reduce adsorption effects. Unfortunately, after approximately 1.5–2 years it appeared the glass diffusion cells placed inside these SilcoNert 1000 treated diffusion chambers started to gain mass, instead of losing mass due to the mercury diffusion. Nevertheless, when connecting the output of a generator to a mercury analyser a very constant concentration of the primary mercury gas standard was produced. Empty glass diffusion cells were placed in a diffusion chamber and also these cells gained mass. It is known mercury is very corrosive, assuming that diffused mercury reacted with the SilcoNert 1000 coating causing the coating to come off the walls and to deposit on the walls of the glass diffusion cells. Therefore, the SilcoNert 1000 treated diffusion chambers were replaced by uncoated stainless steel (AISI 304) diffusion chambers. Mercury will adsorb onto the walls of the chambers but after a while the walls will be saturated with mercury and after this the mercury vapour will not be hampered anymore and the output of the diffusion chambers will be equal to the mercury diffusion rate.

2.5. Bell-jar and Dumarey equation

Currently analysers in the field are calibrated based on the saturation principle, according to ISO 6145-9 [23], using a Bell-jar to generate a saturated mercury concentration in air. The Dumarey equation is most commonly used to calculate the mercury mass concentration in the Bell-jar [7,24,25]. During the study described in this paper a commercially available Bell-jar was used for comparison between the primary mercury gas standard and the Dumarey equation. The Tekran® Model 2505 (Tekran, USA) mercury vapour calibration unit is based on the Bell-jar principle. Since the saturation vapour pressure of mercury is a function of temperature, the exact volume injected, and temperature of the mercury saturated air need to be set in order to determine the mercury injection mass based on the Dumarey equation [7,24,25]. To obtain metrological traceable calibration standards the temperature sensor inside the Bell-jar and the syringe used to transfer the gas mixture were calibrated. The temperature sensor was calibrated between 10 and 25 °C with an expanded uncertainty of 0.005 °C ($k = 2$). The difference between the set temperature and the read out is below the uncertainty, therefore no correction is needed for the temperature read out. To remove volumes of 0.1 mL and 0.2 mL from the Bell-jar a 0.5 mL automated syringe has been used. The syringe was calibrated at 0.1 mL and 0.2 mL. Based on the certification the volume of the syringe should be corrected to 0.100 mL and 0.199 mL, respectively with an expanded uncertainty of 0.007 mL ($k = 2$) for both volumes.

The Dumarey equation was established approximately 30 years ago by Dumarey et al. [7,24,25] and corresponds to the least-squares best fit of results obtained for measurements of the mercury mass concentration in air at saturation (C_{Hg}^{sat}, in ng cm^{-3}) between 288 and 298 K, see Eq. (1).

$$C_{Hg}^{sat} = \frac{D}{T}10^{-\left(A+\left[\frac{B}{T}\right]\right)} \tag{1}$$

T is the temperature of the gas mixture, in K, A is a constant equal to −8.134459741, B is a constant equal to 3240.871534 K, and D is a constant equal to 3216522.61 K ng mL^{-1} [7]. Dumarey et al. estimated that the relative expanded uncertainty ($k = 2$) of data calculated with Eq. (1) is approximately 2% [7]. The standard uncertainty associated with the temperature (0.0025 K) gives rise to an uncertainty contribution to the mercury concentration of 0.0024 ng mL^{-1}, which is negligible in view of the uncertainty of the Dumarey equation itself.

The mass inside the syringe can be calculated by multiplying C_{Hg}^{sat} with the volume taken from the Bell-jar with the syringe (Eq. (1)). Based on the calibration data of the syringe volume (V) the uncertainty budget belonging to the mercury mass (m_{Hg}) is shown in Table 1. The standard uncertainty associated with a mass of 2.22 × 10^{-6} ng can be calculated from Eq. (2) using the law of propagation of uncertainty of the Guide to the expression of Uncertainty in Measurement (GUM) [26]. The expanded uncertainty in this case is 0.09 × 10^{-6} ng ($k = 2$), which is equal to a relative expanded uncertainty of 4%.

$$m_{Hg} = C_{Hg}^{sat}V \tag{2}$$

2.6. Commercial dynamic mercury gas generator

The commercially available mercury vapour generator CavKit 10.534 (PS Analytical (PSA), UK) is a dynamic elemental mercury gas generator based on the continuous generation and dilution of saturated mercury vapour according to ISO 6145-9 [23]. This system is widely used for automated calibration of online mercury analysis. A low gas flow e.g., nitrogen or air (1–20 mL min^{-1} range), controlled by a thermal mass flow-controller passes over a mercury reservoir located in a temperature-controlled oven, picking up a known (calculable) mass-flow of mercury. The mercury mass-flow is then diluted by a gas flow e.g., nitrogen or air from a second thermal mass flow-controller operating in the 2–20 L min^{-1} range. In this approach the expanded uncertainty depends on gas flow rates, temperature and pressure of the mercury source and the vapour pressure equation used. For this work the Dumarey equation was used. A set of equations (Eqs. (1) and (3)) are used to describe the mass of mercury injected by this approach.

$$c_{Hg} = \frac{C_{Hg}^{sat}V_{res}\frac{Tp^{\theta}}{T^{\theta}p}}{V_{res}+V_{dil}} \tag{3}$$

c_{Hg} is the measured concentration of mercury in the gas mixture generated by the dynamic gas generator. This is calculated via Eq. (3) using the standard conditions (293.15 K, 101.3 kPa) reservoir and dilution gas flow rates (V_{res} and V_{dil} respectively), T is the absolute temperature of the mercury reservoir, T^{θ} is the standard reference temperature for the mass flow-controller (293.15 K), p is the measured absolute pressure in the tubing close to the reservoir exit, p^{θ} is the standard reference pressure for the mass flow-controller (101.325 kPa) and C_{Hg}^{sat} the saturated vapour concentration of mercury. C_{Hg}^{sat} is in turn calculated using Eq. (1).

The terms T/T^{θ} and p^{θ}/p in Eq. (3) are corrections of the standard-conditions gas flow, V_{res}, to the actual flow in the heated reservoir. Pressure correction is required to account for both variations in barometric pressure and backpressure applied to the reservoir due to downstream tubing restrictions. The uncertainty budget for the dynamic vapour generator is shown in Table 2. The standard uncertainty associated with a concentration of 9.8 μg m^{-3} can be calculated from equation (3) using the law of propagation of uncertainty according to the GUM [26]. The expanded uncertainty is 0.6 μg m^{-3} ($k = 2$), which is equal to a relative expanded uncertainty of 6%.

2.7. Analysers

For the direct comparison between the primary mercury gas standard and the Dumarey equation a Tekran® 2537B cold vapour atomic fluorescence spectrometer (CVAFS) automated ambient air analyser (Tekran, USA) was used. The instrument samples every 5 min with a flow of 0.5 L min^{-1}, on a gold sorbent trap. Injections from the Tekran® Model 2505 mercury vapour calibration unit are introduced via a home-made sampling port which is connected to the Tekran® 1100 zero air generator (Tekran, USA), providing mercury free air, to direct the sample to the gold trap for sampling. The home-made sampling port consists of a stainless steel Swagelok tube fitting, union Tee 1/8 inch (Swagelok, USA) with a septum for syringe injection on one side, connection to a zero air generator on the other side and the third side is connected to the sampling port of the mercury analyser using 1/8-inch Teflon tubing. The primary mercury gas standard is connected directly to the analyser using 1/8-inch Teflon tubing. This equipment was used in the range from 0.1 μg m^{-3} to 1.0 μg m^{-3}.

A Lumex atomic absorption spectrometer (AAS) utilizing the Zeeman effect (RA-915F, Lumex, Russia) mercury analyser, consisting of a heated cell and a pump, was calibrated using the primary mercury gas

Table 1
Uncertainty budget calculated from equation (2) using the law of propagation of uncertainty according to the GUM for the mercury mass taken from the Bell-jar at 18 °C using a syringe.

Measurand	Value	Distribution	Standard uncertainty	Sensitivity	Uncertainty contribution
C_{Hg}^{sat}	11.14 ng m^{-3}	Normal	0.11 ng m^{-3}	0.199 mL	0.02217 ng
V	0.199 mL	Normal	0.0035 mL	11.14 ng m^{-3}	0.03898 ng
$\mathbf{m_{Hg}}$	**2.22 × 10^{-6} ng**	**Normal**	**0.04 × 10^{-6} ng**		

The bold values are the final values and uncertainties obtained from the other values in the table.

Table 2
Uncertainty budget calculated from equation (3) using the law of propagation of uncertainty according to the GUM for the mercury concentrations from the PSA mercury gas generator.

Measurand	Value	Distribution	Standard uncertainty	Sensitivity coefficient	Uncertainty contribution
C_{Hg}^{sat}	43.2 ng mL^{-1}	Normal	1.4 ng mL^{-1}	0.216 ng^{2} mL^{-2}	0.30 ng mL^{-1}
V_{dil}	15000 mL min^{-1}	Normal	80 mL min^{-1}	−0.00062 ng mL^{-2} min	−0.050 ng mL^{-1}
V_{res}	3.300 mL min^{-1}	Normal	0.033 mL min^{-1}	2.835 ng mL^{-2} min	0.094 ng mL^{-1}
T	308.0 K	Normal	0.5 K	0.034 ng mL^{-1} K	0.017 ng mL^{-1}
p	103.00 kPa	Normal	0.17 kPa	−0.0009 ng mL^{-1} kPa	−0.00014 ng mL^{-1}
c_{Hg}	**9.82 μg m^{-3}**	**Normal**	**0.32 μg m^{-3}**		

The bold values are the final values and uncertainties obtained from the other values in the table.

standard. Subsequently the analyser was used for the certification of the commercially available PSA mercury gas generator. The output of both gas generators is connected directly to the analyser using 1/8-inch Teflon tubing. In contrast to the Tekran® 2537B analyser the Lumex analyser provides a continuous measurement of the gas analysed. This equipment was used in the range from 0.8 to 24 μg m^{-3}.

3. Results and discussion

3.1. Characterisation of diffusion cells

3.1.1. Ø3 mm diffusion cells

To determine the SI-traceable diffusion rate of mercury from the diffusion cells the mass loss is determined by weighing the cells at regular time intervals with a high-resolution and high accuracy balance against an SI-traceable calibrated reference weight. Before the weighing took place, the pressure inside the Ø3 mm diffusion cells was adjusted to 102.00 ± 0.01 kPa by use of the pressure chamber. During a weighing series the cells are closed and the mass difference between the diffusion cell and a reference weight is determined using the AX1006 mass comparator with a Mettler AT1005 balance, inside an airtight container. The weighing series of the 3 cells lasts about 14 days during which the cells are weighed one by one against the same reference weight. For each diffusion cell 80 substitution weighings are performed. During the weighing series a slight increase in mass of the cells is observed that is attributed to the exchange of nitrogen inside the cell with heavier oxygen molecules, as the weighings are performed under air. After the weighing series the oxygen will disappear again after opening the diffusion cells and placing them under nitrogen in the diffusion chamber. The mass difference recordings of the 80 weighings are plotted as a function of time (Fig. 3). A straight line is fitted to determine the mass gain (slope of the straight line) and to enable extrapolation to the time that the diffusion cell was closed (time = 0 days). The regression is

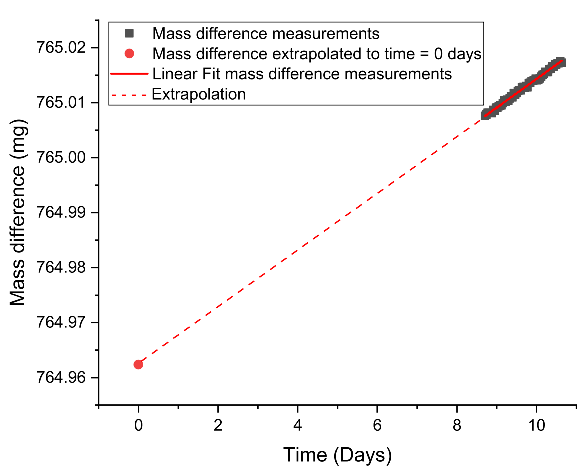

Fig. 3. Extrapolation of the mass difference as a function of time (days) to determine the mass difference (mg) at the moment the cell was closed (time = 0 days).

performed using the errors-in-variables method of ISO 6143 [27], which implements Deming regression [28]. After regression, it was confirmed that the straight line fitted satisfactorily the data. This assessment was performed in accordance with ISO 6143 [27] by checking whether the absolute values of the residuals in x- and y-direction were smaller than the respective expanded uncertainties for each data point. For all datasets, the straight line was sufficient to describe the data.

The mass difference at the moment the cell was closed is determined by extrapolation, using the regression coefficients of the straight line. The uncertainty of the mass difference is calculated using the law of propagation of uncertainty of the GUM [26]. The uncertainty calculations are described in ISO 6143 [27].

Nine weighing series, with in-between periods of several months, of the three Ø3 mm diffusion cells have been performed, with a combined diffusion spanning of 1005 days in total. The mass difference, between the diffusion cells and the reference weight, decreases over time due to the mercury diffusion. In Table 3 and Fig. 4 the mass difference is displayed over time for the mercury diffusion from cell number 20. Time readings are performed when the cell is opened and/or closed, and correspond to the (total) time the diffusion cell was opened in the dynamic mercury gas generator. Opening and closing takes some time and we have modelled that there is some variability in the time measurement hence a standard uncertainty of 5 min was applied for each time the cell was opened and/or closed. Subsequently the uncertainty of the time open (Table 3) was determined by how often the cell was opened and/or closed using the law of propagation of uncertainty according to the GUM [26].

Under steady-state conditions, the diffusion rate is constant in time [16], hence a straight line is the first model to contemplate for describing the data and obtain an estimate of the slope (a_1) (which is the diffusion rate). The underlying physical phenomenon (diffusion through a capillary in the presence of a concentration gradient) is described by Fick's first law of diffusion [29]. As both the time and mass measurements have associated uncertainties, errors in variables regression would be appropriate. Nevertheless, Ordinary Least Squares regression gives comparable diffusion rates and a credible uncertainty [30].

Table 3
Recorded mass differences (mg) as function of time (days and minutes) for diffusion cell number 20 relative to a reference weight. The stated uncertainties are standard uncertainties.

Weighing series	Time open (days)	Time open (min)	Uncertainty time open (min) ($k = 1$)	Mass difference (mg)	Uncertainty mass difference (mg) ($k = 1$)
1	60	86,246	10	766.55250	0.00035
2	130	186,917	12	766.42854	0.00033
3	235	338,083	14	766.26564	0.00038
4	320	460,499	16	766.02791	0.00029
5	407	585,645	17	765.83715	0.00033
6	574	827,023	20	765.52780	0.00057
7	652	939,254	21	765.36752	0.00045
8	873	1,257,486	23	764.96237	0.00094
9	1005	1,447,302	24	764.68905	0.01797

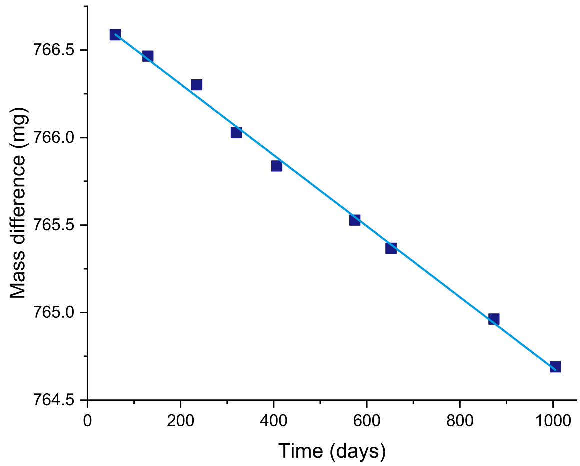

Fig. 4. Graph of the recorded mass differences (mg) as function of time (days) for diffusion cell number 20 relative to a reference weight.

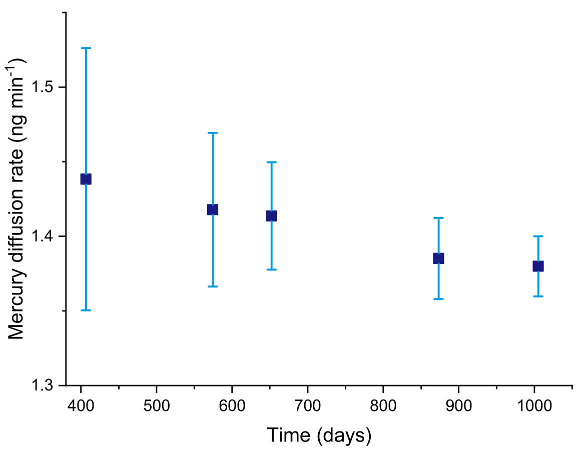

Fig. 5. Mercury diffusion rate (ng min^{-1}) as a function of time (days) determined after weighing series 5 till 9 for diffusion cell number 20. The error bars indicate the expanded uncertainty for the diffusion rate (ng min^{-1}) ($k = 2$).

Model equation $y = a_0 + a_1x$
Regression coefficients:
a_0 (766.679 ± 0.016) mg
a_1 (−1.38 ± 0.02) × 10^{-6} mg min^{-1}

For diffusion cell number 20 a diffusion rate of 1.38 ng min^{-1} is obtained with a standard uncertainty of 0.02 ng min^{-1} which is a relative uncertainty of 1.5%. By performing more weighing series the mercury diffusion rate becomes more accurate and the corresponding uncertainty is becoming smaller (Fig. 5). The diffusion rate was calculated after weighing series 5, 6, 7, 8 and 9 (Table 3). The expanded uncertainty of the diffusion rate after weighing series 5 is 0.18 ng min^{-1} which is decreased to 0.04 ng min^{-1} after weighing series 9. Table 4 shows the diffusion rates and the expanded uncertainties ($k = 2$) of all three Ø3 mm diffusion cells for the last 5 weighing series. The weighing series of diffusion cell 18 after 574 days showed biased results due to the settings of the balance, therefore the result was not considered.

3.1.2. Ø33 mm diffusion cells

Diffusion cells with an opening of Ø33 mm were also developed. These cells have diffusion rates of mercury which are approximately 50 times higher than the Ø3 mm cells. The Ø3 mm cells are interesting to generate gas mixtures containing low mercury concentrations for measurements of mercury in air between 0.1 µg m^{-3} and 2.1 µg m^{-3}. These bigger cells can be used for calibrating equipment used to analyse mercury emissions between 5 µg m^{-3} and 100 µg m^{-3} levels. During a weighing series the cells are closed and the mass difference between the diffusion cell and a reference weight is determined using a Sartorius CC500 analytical balance when the air pressure is between 101.5 kPa and 102.0 kPa. The Ø33 mm diffusion cells have been weighed 11 times, in between the weighings the cells were opened inside the diffusion chambers for a period of 908 days in total. In Fig. 6 and Table 5 the mass difference over time for the mercury diffusion from cell number 35 is shown. Again, the time readings are performed when the cell is opened and/or closed. Opening and closing takes some time and we have modelled that there is some variability in the time measurement hence a standard uncertainty of 5 min was applied for each time the cell was opened and/or closed. Subsequently the uncertainty of the time open (Table 5) was determined by how often the cell was opened and/or closed using the law of propagation of uncertainty according to the GUM [26]. Each weighing series of the Ø33 mm cells only lasts a few hours instead of two weeks (as for the Ø3 mm diffusion cells), therefore, the mass difference determined during the weighing is used. The uncertainty of the conventional mass is determined considering the uncertainty sources of type A evaluation; the reference weight, balance resolution, air buoyancy and reproducibility.

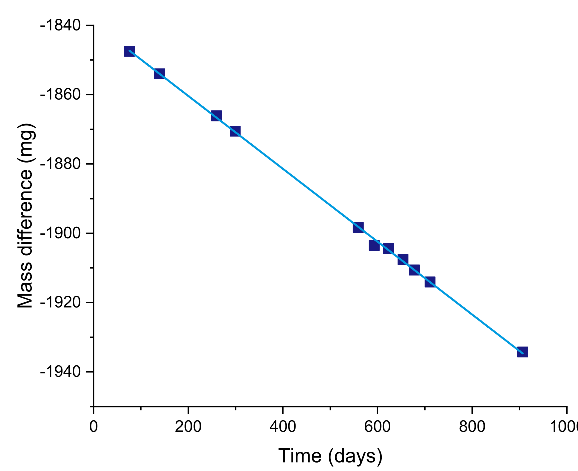

Fig. 6. Graph of the recorded mass differences (mg) as function of time (days) for diffusion cell number 35 relative to a reference weight.

Table 4
Mercury diffusion rates (ng min^{-1}) as a function of time (days) of the three Ø3 mm diffusion cells, with numbers 18, 19 and 20. In parentheses the expanded uncertainty for the diffusion rate (ng min^{-1}) ($k = 2$).

Time the diffusion cells were open	Diffusion rate (ng min^{-1})				
	407 days	574 days	652 days	873 days	1005 days
Cell 18	1.45 (0.06)		1.37 (0.06)	1.30 (0.06)	1.29 (0.06)
Cell 19	1.48 (0.20)	1.43 (0.12)	1.44 (0.08)	1.41 (0.06)	1.51 (0.10)
Cell 20	1.44 (0.18)	1.42 (0.10)	1.41 (0.08)	1.39 (0.06)	1.38 (0.04)

Table 5
Recorded mass differences (mg) as a function of time (days and minutes) for diffusion cell number 35 relative to a reference weight. The stated uncertainties are standard uncertainties.

Weighing series	Time open (days)	Time open (min)	Uncertainty time open (min) (k = 1)	Mass difference (mg)	Uncertainty mass difference (mg) (k = 1)
1	76	109,380	10	−1795.18321	0.0678
2	140	201,245	12	−1799.03013	0.0803
3	260	373,965	14	−1813.15487	0.1092
4	299	431,175	16	−1816.69987	0.1016
5	559	805,575	17	−1843.57333	0.1063
6	593	854,190	19	−1847.60333	0.1229
7	623	897,335	20	−1850.79757	0.1043
8	654	941,745	21	−1853.95987	0.1119
9	678	976,240	22	−1855.37795	0.1148
10	711	1,023,965	23	−1859.69718	0.1110
11	907	1,305,805	24	−1880.65564	0.1285

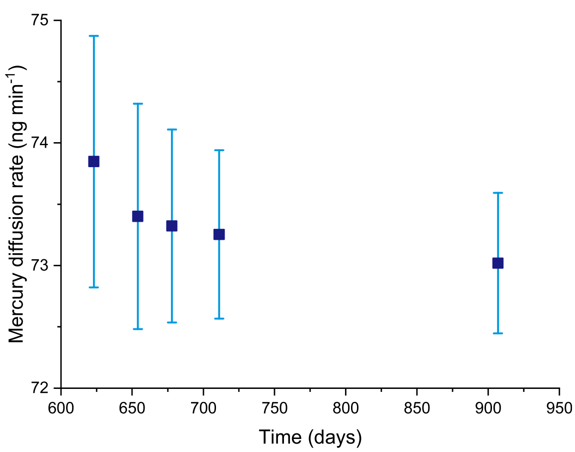

Fig. 7. Mercury diffusion rate (ng min^{-1}) as a function of time (days) determined after weighing series 5 till 9 for diffusion cell number 35. The error bars indicate the expanded uncertainty for the diffusion rate (ng min^{-1}) (k = 2).

When the data points are fitted in a straight line, using Ordinary Least Squares regression, the following regression coefficients for the intercept and the slope have been obtained [30].

Model equation $y = a_0 + a_1x$
Regression coefficients:
a_0 (−1839.3 ± 0.5) mg
a_1 (−7.30 ± 0.06) × 10^{-5} mg min^{-1}

The slope is the diffusion rate of mercury. For cell 35 a diffusion rate of 73.0 ng min^{-1} is obtained with a standard uncertainty of 0.6 ng min^{-1} which is a relative uncertainty of 0.9%. The Ø33 mm cells have a mercury diffusion rate which is approximately 50 times higher than the Ø3 mm cells. Therefore, it is easier to determine the mass loss and a smaller relative standard uncertainty has been obtained for the Ø33 mm cell (0.9%) compared to the relative standard uncertainty of the Ø3 mm cell (1.5%). Once more, by performing more weighing series the mercury diffusion rate becomes more accurate and the uncertainty smaller (Fig. 7). The diffusion rate has been calculated after weighing series 7, 8, 9, 10 and 11 (Table 5). The expanded uncertainty of the diffusion rate after weighing series 7 is 2.0 ng min^{-1} which is decreased to 1.2 ng min^{-1} after weighing series 11. Table 6 shows the diffusion rates and the expanded uncertainties (k = 2) of all three Ø33 mm cells for the last 5 weighing series. Cell 34 has not been weighed as many times as cells 35 and 36 therefore only the last 4 weighing series are shown, while the uncertainty of the obtained mercury diffusion rate is much higher.

Table 6
Mercury diffusion rates (ng min^{-1}) as a function of time (days) of the three Ø33 mm diffusion cells, with numbers 34, 35 and 36. In parentheses the expanded uncertainty for the diffusion rate (ng min^{-1}) (k = 2).

Time the diffusion cells were open	Diffusion rate (ng min^{-1})				
		95 days	119 days	152 days	348 days
Cell 34		54 (17)	61 (12)	70 (12)	72 (4)
	623 days	654 days	678 days	711 days	907 days
Cell 35	73.8 (2.0)	73.4 (1.8)	73.3 (1.6)	73.3 (1.4)	73.0 (1.2)
Cell 36	71.2 (2.0)	71.4 (1.6)	71.4 (1.6)	71.2 (1.4)	72.1 (1.2)

Table 7
Uncertainty budget calculated from equation (4) using the law of propagation of uncertainty according to the GUM for the primary mercury gas standard concentrations.

Measurand	Value	Distribution	Standard uncertainty	Sensitivity	Uncertainty contribution
$\overline{q}_m$(Hg cell 18)	1.29 ng min^{-1}	Normal	0.03 ng min^{-1}	0.106 min L^{-1}	0.00267 ng L^{-1}
$\overline{q}_m$(Hg cell 19)	1.51 ng min^{-1}	Normal	0.05 ng min^{-1}	0.106 min L^{-1}	0.00581 ng L^{-1}
$\overline{q}_m$(Hg cell 20)	1.38 ng min^{-1}	Normal	0.02 ng min^{-1}	0.106 min L^{-1}	0.00213 ng L^{-1}
$q_v(air)$	9.45 L min^{-1}	Normal	0.05 L min^{-1}	−0.047 ng L^{-2} min	−0.00215 ng L^{-1}
p	103.0 kPa	Normal	0.05 kPa	−0.0434 ng L^{-1} kPa	−0.00022 ng L^{-1}
$\boldsymbol{C_{Hg}}$	**0.441 µg m^{-3}**	**Normal**	**0.007 µg m^{-3}**		

The bold values are the final values and uncertainties obtained from the other values in the table.

3.2. Direct comparison between the primary mercury gas standard and the Dumarey equation

Using the primary mercury gas standard and the Bell-jar, gas mixtures have been obtained in the range 0.3–0.9 µg m^{-3}, to perform a direct comparison between these two methods. The mercury concentrations obtained using the primary mercury gas standard have been calculated using equation (4).

$$c_{Hg} = \frac{\overline{q}_m(Hg)}{q_v(air)}\frac{p^\theta}{p} \tag{4}$$

c_{Hg} is the output concentration of the dynamic gas generator, $\overline{q}_m$(Hg) is the mercury diffusion rate, if multiple cells are used the sum of the diffusion rates is used. q_v(air) is the total gas flow; the nitrogen flow from the diffusion chamber and the air flow used to dilute the mercury vapours to the desired concentration. p^θ/p in Eq. (4) is a correction to the standard ambient pressure (101.325 kPa). For this comparison the Ø3 mm diffusion cells are used. Mercury concentrations were obtained by using two Ø3 mm cells or using all three Ø3 mm cells. The standard uncertainty associated with a mercury concentration of 0.441 µg m^{-3} obtained using all three Ø3 mm cells was calculated from equation (4) using the law of propagation of uncertainty according to the GUM [26] (Table 7). The temperature is not included in the uncertainty as the temperature of the system is maintained at room temperature (20.0 ± 0.1 °C). A temperature variation of 0.1 °C leads to a deviation in the mercury diffusion rate of 0.03%. This is a factor 100 below the relative expanded uncertainty of 3%. The primary mercury gas standard can be used to generate mercury concentrations with a lower uncertainty compared to the empirical equations. Reference materials currently used in the field have relative uncertainties ranging from 5% up to 21% [9–13].

For this comparison the temperature of the Bell-jar has been set to 16.0 °C or 18.0 °C and the mercury concentration of the gas mixtures obtained with the Bell-jar is calculated using the Dumarey equation (Eq. (1)). Gas mixtures, with a similar concentration, obtained with both methods were cross-checked by means of the Tekran CVAFS mercury analyser (Table 8). During the comparison both gas mixtures were analysed 10 times in a row. The average results obtained on the same day are compared with each other. The measurements of the primary standard are used as one-point calibration. The measurement from the Bell-jar was compared to the calibration. This way the relative difference has been determined between the concentrations obtained from the primary mercury gas standard and the Bell-jar in combination with the Dumarey equation (Table 8 and Fig. 8). The average relative difference of the 6 entries is 3.8% with a standard deviation of 0.8%. This means that the output of the Bell-jar is 3.8% below the concentration of the primary mercury gas standard.

3.3. Mercury analyser calibration and mercury generator certification

Proper calibration of instruments and methods for mercury measurements is crucial to establish metrological traceability to the SI units.

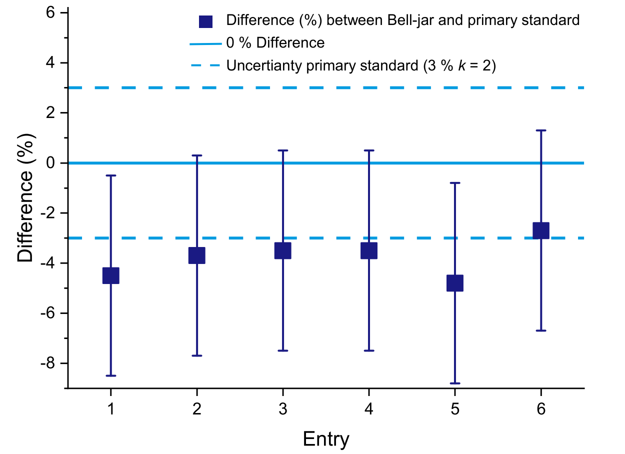

Fig. 8. Relative differences and expanded uncertainties determined during the direct comparison between the primary mercury gas standard and the Bell-jar in combination with the Dumarey equation.

Table 9
Mercury concentrations obtained with three Ø3 mm cells and one Ø33 mm cell and the corresponding responses obtained during the calibration of the analyser in the range of 0.8–24 µg m^{-3}, including the corresponding expanded uncertainty (U (k = 2)) for the obtained concentration and standard deviation (std. dev.) for the response in parentheses.

Diffusion cells	c_{Hg}(µg m^{-3}) (U (k = 2))	Response (µg m^{-3}) (*std. dev.*)
3 × Ø3 mm	0.821 (0.024)	0.78 (0.02)
3 × Ø3 mm	1.36 (0.04)	1.28 (0.03)
3 × Ø3 mm	1.37 (0.04)	1.30 (0.02)
1 × Ø33 mm	4.75 (0.08)	4.41 (0.02)
1 × Ø33 mm	6.45 (0.012)	6.02 (0.03)
1 × Ø33 mm	10.26 (0.18)	9.65 (0.03)
1 × Ø33 mm	14.65 (0.26)	13.66 (0.03)
1 × Ø33 mm	16.27 (0.30)	15.26 (0.04)
1 × Ø33 mm	24.60 (0.44)	23.10 (0.04)
1 × Ø33 mm	14.58 (0.26)	13.73 (0.03)
1 × Ø33 mm	4.75 (0.08)	4.44 (0.03)
3 × Ø3 mm	1.36 (0.04)	1.31 (0.02)

The primary mercury gas standard was used to calibrate a Lumex AAS analyser utilizing the Zeeman effect (RA-915F) in the range 0.8–24 µg m^{-3} for stack gas emission measurements. Subsequently, the calibrated analyser was used for the certification of the PSA Cavkit dynamic mercury gas generator. These generators are capable of generating calibration gas mixtures for use in both laboratory and field conditions.

For the calibration of the equipment both the Ø3 mm and the Ø33 mm diffusion cells were used, four mercury concentrations were obtained with the three Ø3 mm cells and seven with Ø33 mm cell number 36 (Table 9). The relative expanded uncertainty for the concentrations

Table 8
The relative differences determined during the direct comparison between the primary mercury gas standard and the Bell-jar in combination with the Dumarey equation. For the comparison 6 mercury concentrations have been obtained using different settings of the primary mercury gas standard (diffusion rate and air flow rate) and the Bell-jar (temperature and injected volume).

Entry	Primary mercury gas standard			Bell-jar – Dumarey equation			Difference (%)
	$\overline{q}_m$(Hg)(ng min^{-1})	$q_m(air)$(L min^{-1})	c_{Hg}(µg m^{-3}) (U (k = 2))	Temperature (°C)	V (syringe)(mL)	c_{Hg}(µg m^{-3}) (U (k = 2))	
1	2.89	7.4808	0.386 (0.016)	16.0	0.100	0.376 (0.015)	−4.5
2	2.89	6.3258	0.457 (0.019)	18.0	0.100	0.446 (0.018)	−3.7
3	4.18	11.1836	0.374 (0.012)	16.0	0.100	0.376 (0.015)	−3.5
4	4.18	9.4764	0.411 (0.014)	18.0	0.100	0.446 (0.018)	−3.5
5	4.18	5.6254	0.743 (0.023)	16.0	0.199	0.752 (0.030)	−4.8
6	4.18	4.7651	0.877 (0.027)	18.0	0.199	0.891 (0.036)	−2.7

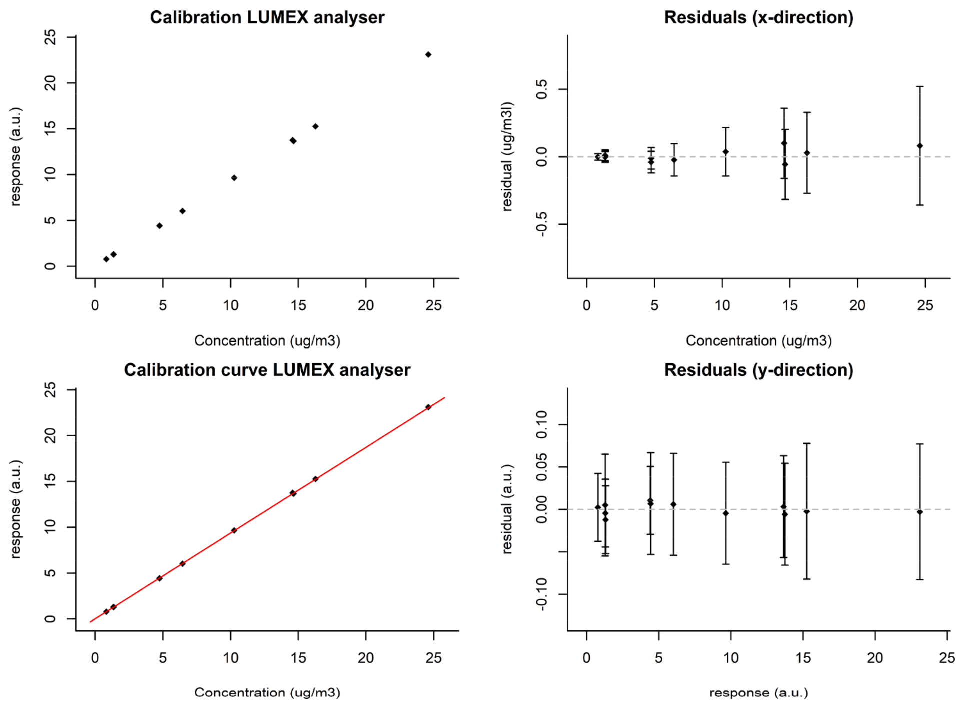

Fig. 9. Representation of the calibration results of the Lumex analyser calculated according to ISO 6143. Top left: calibration data, bottom left: calibration data and curve, top right: residuals in the concentration, bottom right: residuals in the response.

Table 10
The relative differences determined between the setpoint of the mercury gas generator and the certified value. The uncertainties in parentheses are expanded uncertainties.

Setpoint c_{Hg} (µg m^{-3}) (U (k = 2))	Certification c_{Hg} (µg m^{-3}) (U (k = 2))	Difference (%)
10.0 (0.6)	9.63 (0.30)	−3.5
13.6 (0.8)	13.4 (0.4)	−0.4
18.1 (1.1)	18.2 (0.4)	0.5
22.3 (1.3)	22.8 (0.6)	0.6

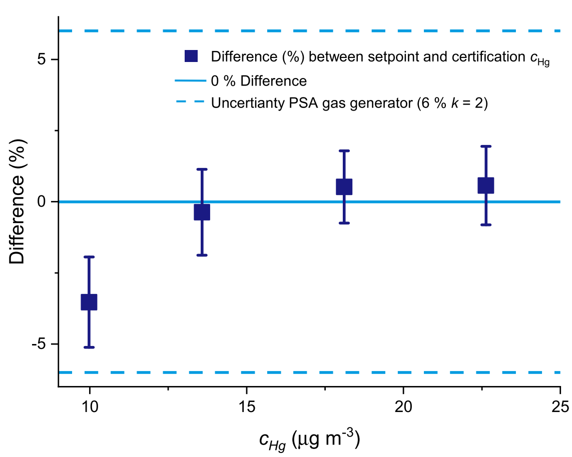

Fig. 10. Graphic representation of the relative differences and expanded uncertainties determined between the setpoint of the mercury gas generator and the certified value.

obtained with the Ø3 mm cells is 3% and for the Ø33 mm it is 1.8% both calculated as described before (Eq. (4) and Table 7). The response of the analyser is given in µg m^{-3}.

The data points of the calibration have been fitted in a straight line according to ISO 6143 (Fig. 9) [27]. The following regression coefficients for the intercept and the slope were obtained.

Model equation $y = a_0 + a_1 x$
Regression coefficients:
a_0 (0.015 ± 0.015) µg m^{-3}
a_1 (0.935 ± 0.004) µg m^{-3}

After the calibration of the Lumex detector the output of the PSA mercury gas generator was certified. The output of the generator was connected to the analyser and four different mercury concentrations were generated. Using the regression coefficients obtained during the calibration the gas generator has been certified according to ISO 6143 and the difference between the calibration and the output of the dynamic gas generator has been determined (Table 10 and Fig. 10) [27]. The certification values fall within the uncertainty of the dynamic gas generator (Table 2) and deviate less than 1% from the primary mercury gas standard for the three higher concentrations.

4. Conclusion

The primary mercury gas standard has been improved to obtain SI-traceability for mercury measurement results. Using the previously developed advanced mercury loss weighing process the novel stainless steel diffusion cells have been fully characterised. Mercury concentrations can be obtained between 0.1 µg m^{-3} and 2.1 µg m^{-3} with a relative expanded uncertainty of 3% and between 5 µg m^{-3} and 100 µg m^{-3} with a relative expanded uncertainty of 1.8% (k = 2).

In the field mercury concentration measurements rely on the Dumarey vapour pressure equation. Based on the comparisons we performed a bias of 3.8% was observed relative to the primary method.

The primary standard considerably strengthens the metrological traceability chain for mercury concentration at emission and ambient levels. This strengthening is essential to control and assess mercury concentrations in the environment.

CRediT authorship contribution statement

Iris de Krom: Methodology, Investigation, Writing - original draft, Visualization, Supervision. **Wijnand Bavius:** Validation. **Ruben Ziel:** Validation. **Evtim Efremov:** Validation. **Davina van Meer:** Validation. **Peter van Otterloo:** Validation. **Inge van Andel:** Methodology, Investigation. **Deborah van Osselen:** Validation. **Maurice Heemskerk:** Methodology, Resources. **Adriaan M.H. van der Veen:** Methodology, Writing - review & editing. **Matthew A. Dexter:** Validation, Investigation, Resources. **Warren T. Corns:** Validation, Investigation, Resources, Writing - review & editing. **Hugo Ent:** Conceptualization, Methodology, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Global mercury assessment 2018, 2020. https://wedocs.unep.org/bitstream/handle/20.500.11822/27579/GMA2018.pdf?sequence=1&isAllowed=y and https://www.unenvironment.org/resources/publication/global-mercury-assessment-2018 (accessed 1 April 2020).
[2] European Directive 2010/75/EU on industrial emissions (integrated pollution prevention and control), 2020. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32010L0075 (accessed 1 April 2020).
[3] European Directive 2004/107/EC relating to arsenic, cadmium, mercury, nickel and polycyclic aromatic hydrocarbons in ambient air, 2020. https://eur-lex.europa.eu/eli/dir/2004/107/oj (accessed 1 April 2020).
[4] European Directive 2000/76/EC on the incineration of waste, 2020. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32000L0076 (accessed 1 April 2020).
[5] Minamata Convention, 2020. http://www.mercuryconvention.org/ (accessed 1 April 2020).
[6] ISO/IEC 17025:2017: General requirements for the competence of testing and calibration laboratories, 2020. https://www.iso.org/obp/ui/#iso:std:iso-iec:17025:ed-3:v1:en (accessed 1 April 2020).
[7] R. Dumarey, R.J.C. Brown, W.T. Corn, A.S. Brown, P.B. Stockwell, Accred Qual. Assur. 15 (2010) 409, https://doi.org/10.1007/s00769-010-0645-1.
[8] M.L. Huber, A. Laesecke, D.G. Friend, Ind. Eng. Chem. Res. 45 (2006) 7351, https://doi.org/10.1021/ie060560s.
[9] A.S. Brown, R.J.C. Brown, R.E. Yardley, W.T. Corns, P.B. Stockwell, Atmos. Environ. 42 (2008) 2504, https://doi.org/10.1016/j.atmosenv.2007.12.012.
[10] A.S. Brown, R.J.C. Brown, W.T. Corns, P.B. Stockwell, Analyst 133 (2008) 946, https://doi.org/10.1039/B803724H.
[11] A.S. Brown, R.J.C. Brown, M.A. Dexter, W.T. Corns, P.B. Stockwell, Anal. Meth. 2 (2010) 95, https://doi.org/10.1039/C0AY00058B.
[12] R.J.C. Brown, A.S. Brown, Analyst 133 (2008) 1611, https://doi.org/10.1039/b806860g.
[13] R.J.C. Brown, A.S. Brown, W.T. Corns, P.B. Stockwell, Instr. Sci. Tech. 6 (2008) 611, https://doi.org/10.1080/10739140802448309.
[14] S.E. Long, J.E. Norris, J. Carney, J.V. Ryan, Atm. Pol. Res. (2020), https://doi.org/10.1016/j.apr.2020.02.003 in press.
[15] H. Ent, I. van Andel, M. Heemskerk, R.P. Otterloo, W. Bavius, A. Baldan, M. Horvat, R.J.C. Brown, C.R. Quétel, Meas. Sci. Technol. 25 (2014), 115801, https://doi.org/10.1088/0957-0233/25/11/115801.
[16] ISO 6145-8:2005 – Gas analysis – Preparation of calibration gas mixtures using dynamic volumetric methods – Part 8: Diffusion method. Geneva, Switzerland, 2020. https://www.iso.org/standard/36480.html (accessed 1 April 2020).
[17] BIPM, IEC, IFCC, ILAC, ISO, IUPAC and OIML, 2012, International vocabulary of metrology – Basic and general concepts and associated terms, third edition, BIPM, 2012.
[18] W. Richter, Accred Qual. Assur. 2 (1997) 354, https://doi.org/10.1007/s007690050165.
[19] M.J.T. Milton, T.J. Quinn, Metrologia 38 (2001) 289, https://doi.org/10.1088/0026-1394/38/4/1.
[20] P. Taylor, H. Kipphardt, P. De Bièvre, Accred Qual. Assur. 6 (2001) 103, https://doi.org/10.1007/PL00010444.
[21] I. de Krom, W. Bavius, R. Ziel, E.A. McGhee, R.J.C. Brown, I. Živković, J. Gačnik, V. Fajon, J. Kotnik, M. Horvat, H. Ent, Comparability of calibration strategies for measuring mercury concentrations in gas emission sources and the atmosphere, 2020 unpublished results.
[22] Organisation Internationale de Métrologie Légale, OIML R 111-1 Weights of classes E1, E2, F1, F2, M1, M1–2, M2, M2–3 and M3. Part 1: Metrological and technical requirements, 2004, Paris, France.
[23] ISO 6145-9:2009 – Gas analysis – Preparation of calibration gas mixtures using dynamic volumetric methods – Part 9: Saturation method. Geneva, Switzerland, 2020. https://www.iso.org/standard/45469.html (accessed 1 April 2020).
[24] R. Dumarey, R. Heindryckx, R. Dams, J. Hoste, Anal. Chim. Acta 107 (1979) 159, https://doi.org/10.1016/S0003-2670(01)93206-4.
[25] R. Dumarey, E. Temmerman, R. Dams, J. Hoste, Anal. Chim. Acta 170 (1985) 337, https://doi.org/10.1016/S0003-2670(00)81759-6.
[26] BIPM, IEC, IFCC, ILAC, ISO, IUPAC and OIML, 2008, Guide to the expression of uncertainty in measurement JCGM 100:2008, GUM 1995 with Minor Corrections (BIPM), 2020. https://www.bipm.org/utils/common/documents/jcgm/JCGM_100_2008_E.pdf (accessed 1 April 2020).
[27] ISO 6143:2001 – Gas analysis – Comparison methods for determining and checking the composition of calibration gas mixtures. Geneva, Switzerland, 2020. https://www.iso.org/standard/24665.html (accessed 1 April 2020).
[28] W.E. Deming, Statistical Adjustment of Data. Dover books on elementary and intermediate mathematics, J. Wiley & Sons, Incorporated, 1943.
[29] P. Atkins, J. de Paula, Physical Chemistry, eighth ed., W.H Freeman, 2006.
[30] N.R. Draper, H. Smith, Applied Regression Analysis, second ed., J. Wiley and Sons, New York, 1981.